

V1 student actor path

Actor-visible state

$s_t^{\text{cf0}}, \varphi_e$

scene memory, history

Q_t , masks

**Current-belief
products**

optional belief render
or support/ray summaries
from actor memory only

Student value

$Q_{H,\theta}(s_t^{\text{cf0}}, e, q_{t,i})$

valid-row scores

Privileged teacher / oracle

GT assets

mesh, GT target crop

GT boxes for matching

**Dense candidate
render**

depth, valid mask,
candidate points, target error

Oracle values

$r_t^e, G_t^{(H)}$

upper bounds or teacher targets

Training-only signals

Return / ranking loss

Monte-Carlo or TD target
masked over $m_{t,i} = 1$

Optional distillation

stop-gradient teacher value
ablation, not actor input

forbidden at V1 inference:
GT depth, GT crop, or dense
candidate render as actor features